Experiment → 2

Aim → To convert an ER diagram into a relational schema and implement it in MYSQL using primary keys, Foreign key, NOT NULL, Unique, ON DELETE CASCADE, and ON DELETE SET NULL constraints. Also, to insert sample data and demonstrate referential integrity violation.

Source Code

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;
CREATE TABLE CUSTOMER(
  CustomerID INT PRIMARY KEY,
  Name VARCHAR(50) NOT NULL,
  EMAIL VARCHAR(100) NOT NULL UNIQUE,
  Password VARCHAR(100) NOT NULL,
  DATE OF BIRTH DATE NOT NULL
  );

  CREATE TABLE CUSTOMER_PHONE(
    CustomerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY(customerID, Phone),
    FOREIGN KEY(CustomerID) REFERENCE
    Customer(CustomerID)
    ON DELETE CASCADE
  );
```

```
CREATE TABLE CUSTOMER_ALT_PHONE(
    CustomerID INT
    ALTPhone VARCHAR (15),
    PRIMARY KEY (CustomerID , ALTPhone),
    FOREIGN KEY(CustomerID) REFERENCE customer(
      (CustomerID)
   ON DELETE CASCADE
   );

CREATE TABLE SELLER(
   SELLERID  INT PRIMARY KEY,
   Seller Name VARCHAR (50) NOT NULL,
   Email VARCHAR (100) UNIQUE NOT NULL,
   GSTIN VARCHAR (20) UNIQUE NOT NULL,
   Account NO VARCHAR (15),
   IFSCcode VARCHAR (15),
   Bank Name VARCHAR (20)
   );

CREATE TABLE SELLER_PHONE(
    SellerID INT,
    Phone VARCHAR (15),
    PRIMARY KEY (SellerID, Phone),
    FOREIGN KEY(SellerID) REFERENCE SELLER(SellerID)
    ON DELETE CASCADE
   );

CREATE TABLE CATEGORY(
  CategoryID INT PRIMARY KEY,
  Category Name VARCHAR (50) NOT NULL,
  Description VARCHAR (100),
  Parent Category ID INT,
  FOREIGN KEY (Parent categoryID) REFFERENCE CATEGORY
   (CategoryID)
   ON DELETE SET NULL
);
```

```
CREATE TABLE PRODUCT (
 ProductID INT PRIMARY KEY,
ProductName VARCHAR(50) NOT NULL,
Description VARCHAR(100),
Price Decimal (10,2), NOT NULL,
StockQty INT NOT NULL,
Weight DECIMAL(8,2),
SellerID INT NOT NULL,
CategoryID INT,
FOREIGN KEY(SellerID) REFFERENCE SELLER(SellerID)
ON DELETE CASCADE
FOREIGN KEY (CategoryID) REFFERENCE CATEGORY
 (CategoryID)
ON DELETE SET NULL
);

CREATE TABLE PRODUCT-IMAGE(
 ProductID INT PRIMARY KEY,
 Bond VARCHAR(50),
 Warranty VARCHAR(30),
 PowerRating VARCHAR(30),
 FOREIGN KEY (ProductID) REFFERENCE Product(ProductID)
 ON DELETE CASCADE
 );
CREATE TABLE CLOTHING(
 ProductID INT PRIMARY KEY,
 SIZE VARCHAR(20),
 Colour VARCHAR(20),
 Material VARCHAR(30),
 FOREIGN KEY (ProductID) REFFERENCE PRODUCT (ProductID)
 ON DELETE CASCADE
 );

CREATE TABLE BOOKS(
 ProductID INT PRIMARY KEY,
 ISBN VARCHAR(20) UNIQUE,
 Author VARCHAR(50),
 Publisher VARCHAR(50),
 FOREIGN KEY (ProductID) REFFERENCE PRODUCT (ProductID)
  ON DELETE CASCADE);
```

```
CREATE TABLE ADDRESS(
 AddressID INT PRIMARY KEY,
 Door NO VARCHAR (10),
 Street VARCHAR (50),
 Area VARCHAR (50),
 City VARCHAR (30),
 State VARCHAR (30),
 PINCODE VARCHAR (10) NOT NULL,
 Address type VARCHAR (20),
 CustomerID INT NOT NULL,
 FOREIGN KEY(CustomerID) REFFERENCE CUSTOMER(CustomerI
 ON DELETE CASCADE
 );

CREATE TABLE ADDRESS_MOBILE(
 AddressID INT,
 mobileNO VARCHAR (15),
 PRIMARY KEY (AddressID, mobileNO),
 FOREIGN KEY(AddressID) REFFERENCE Address(AddressID)
 ON DELETE CASCADE
 );

 CREATE TABLE ORDER(
 OrderID INT PRIMARY KEY,
 OrderDate DATE NOT NULL,
 Order Status VARCHAR (20) NOT NULL,
 Total amount DECIMAL (10,2) NOT NULL,
 CustomerID INT NOT NULL,
 AddressID INT,
 FOREIGN KEY (CustomerID) REFFERENCE CUSTOMER (CustomerID
 ON DELETE CASCADE,
 FOREIGN KEY (AddressID) REFFERENCE ADDRESS (AddressID)
 ON DELETE SET NULL
 );
CREATE TABLE ORDER_COUPON(
 Order ID INT,
 Coupon Code VARCHAR (30)
 PRIMARY KEY (OrderID, Coupon code),
 FOREIGN KEY (OrderID) REFFERENCE ORDER (OrderID)
  ON DELETE CASCADE
 );
```

```
CREATE TABLE ORDER ITEM(
  OrderID INT,
  ItemNO INT,
  ProductID INT NOT NULL,
  ItemPrice DECIMAL (10,2) NOT NULL,
  Quantity INT NOT NULL),
  Discount DECIMAL (10,2),
  TAX Amount DECIMAL (10,2),
  PRIMARY KEY (OrderID, ITEMNO),
  FOREIGN KEY (orderID) REFFERENCE ORDER(OrderID)
  ON DELETE CASCADE,
FOREIGN KEY( ProductID), REFFERENCE PRODUCT (ProductID)
  ON DELETE CASCADE
  );
CREATE TABLE PAYMENT (
  Payment ID INT PRIMARY KEY,
  Payment Date DATE NOT NULL,
  Amount DECIMAL(10,2) NOT NULL,
  Payment method VARCHAR (30) NOT NULL,
  TransactionID VARCHAR (30) UNIQUE,
  Payment Status VARCHAR(20) NOT NULL,
  OrderID INT UNIQUE NOT NULL,
FOREIGN KEY (orderID), REFFERENCE ORDER (orderID)
  ON DELETE CASCADE
  );
CREATE TABLE DELIVERY(
  DeliveryID INT PRIMARY KEY,
  Delivery Date DATE,
  Delivery Status VARCHAR (30) NOT NULL,
  Tracking ID VARCHAR (30) UNIQUE,
  Delivery Partner VARCHAR (50),
  Shipping Partner VARCHAR (50),
  Shipping charge DECIMAL (8,2),
  ORDER ID INT UNIQUE NUT NULL,
  Address ID INT,
  FOREIGN KEY (order) REFFERENCE ORDER (orderID)
  ON DELETE CASCADE,
  FOREIGN KEY (Address ID) REFERENCE ADDRESS(AddressID)
  ON DELETE SET NULL
  );
```

```
INSERT INTO CUSTOMER VALUE:
(1, 'Adarsh', 'adarsh8876@gmail.com', 'Pass123' 2006-01-09

INSERT INTO CUSTOMER_PHONE VALUE
(1, '8953318721);

INSERT INTO SELLER VALUE
(101, 'Techstore', 'tech@gmail.com', 'GST123',
'123456', SBI4001, 'SBI');

INSERT INTO SELLER-PHONE VALUE
(101, '9123456789');

INSERT INTO CATEGORY VALUE
(201, 'Electronics', 'Electronic item', NULL);

INSERT INTO CATEGORY VALUE
(202, 'mobile', 'mobile-Phones', 201);

INSERT INTO PRODUCT VALUES
(301, 'Laptop', 'Gaming Laptop', 75000, 10, 2.5, 101, 201);

INSERT INTO ELECTRONICS VALUES
(301, 'Lenovo', 2years, '6SW');

INSERT INTO PRODUCT-IMAGE VALUES
(301, 'Laptop.jpg');

INSERT INTO ADDRESS VALUES
(401, '10', 'meera Bihar colony', 'Transport nagar',
'Lucknow', 'UttarPradesh', '226014', 'Home');

INSERT INTO ADDRESS_MOBILE VALUE
(401, '9853318721');

INSERT INTO ORDER VALUE
(501, '2026-08-14', 'Placed', 75000, 1, 401);

INSERT INTO ORDERITEM VALUE
(501, 1, 301, 75000, 1, 1000, 9000);
```

```
INSERT INTO ORDER_COUPON VALUE
(501, 'Coupons');

INSERT INTO PAYMENT VALUES
(601, '2025-08-14', '74000', 'UPI :TxN123, "Paid");

INSERT INTO DELIVERY VALUES
(701, Null, 'Processing', 'TRK123', 'Delivery', 100,
501, 401);

-- Refe REFFERENTIAL INTEGRITY VIOLATION
  INSERT INTO ORDERS VALUES
(502, '2026-08-14', 'Placed', 5000, 999, 401);

-- SELL NULL DEMONTRATION
  DELETE FROM CATEGORY WHERE CategoryID = 201;

-- CASCADE DEMONSTRATION
  DELETE FROM Customer WHERE CustomerID = 1;
```

Output → 1

CustomerID	CustomerName	Email	Password	DOB
1	Adarsh	adarsh8876@gmail.com	Pass123	2006-04-09

CustomerID	Phone NO
1	8953318721

CustomerID	Alt Phone
1	7307598187

SellerID	SellerName	Email	GSTNO	IFSC	BANKNO
101	Techstore	Tech@gmail.com	GST123	SBI	SBI0001	123456

SellerID	Phone
101	9812345678

CategoryID	CategoryName	Description	Parent CategoryID
201	Electronics	Electronic item	NULL
202	mobiles	mobile phone	101

ProductID	Product Name	Description	Price	Stock	Weight	SellerID	CategoryID
301	Laptop	Gaming Laptop	75000.00	10	2.50	101	201

ProductID	Image
301	Laptop.JPG

ProductID	Brand	Warranty	PowerRating
301	Lenovo	2years	65W

AddressID	DoorNO	Street	Area	City	State	Pincode	Addstype	CustomerID
401	12	meera Dinar colony	Transport Nagar	Lucknow	UP	226014	Home	1

AddressID	mobileNO
401	8953319721

OrderID	order Date	order status	Total amount	Customer ID	Address ID
501	2020-08-14	Placed	7500.00	1	401

orderID	~~Coupon~~ ITEMNO	Discount	ProductID	Item Price	Quality	Tax amount
501	1	1000.00	301	7500	1	9000.00

OrderID	Coupon code
501	Coup123

PaymentID	Payment date	Amount	Payment method	Transaction ID	Payment Status	orderID
601	2026-08-14	74000.00	UPI	TXN123	PAID	501

Delivery ID	Delivery date	Delivery Status	Tracking ID	Delivery Partner	Shipping charge	order ID	Address ID
701	Null	Processing	TRK123	Delivery	100.00	501	401